ICT2113 · Data Structures and Algorithms

Foundations: Data, Algorithms, and Cost

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

An algorithm is a finite, precise sequence of steps that transforms inputs into outputs. Data structures organize values so operations such as lookup, insertion, and deletion have predictable costs. An abstract data type describes allowed operations independently of the representation.

2. Example and reasoning

For an array-backed list, indexed reads are constant time, while inserting at the front shifts later elements. A linked list changes the trade-off: it can splice nodes once a position is known, but finding that position requires traversal.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “foundations: data, algorithms, and cost” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.